Operating Systems - Unit 3: Deadlocks

A deadlock is a situation where a set of processes are blocked, because each process is holding a resource and waiting for another resource held by some other process.

Four necessary conditions (Coffman conditions):
1. Mutual exclusion: a resource can be used by only one process.
2. Hold and wait: a process holds one resource and waits for more.
3. No preemption: a resource cannot be taken away by force.
4. Circular wait: processes wait for each other in a cycle.

Handling deadlocks

Prevention: make sure at least one condition can never hold.
Avoidance: grant a request only if the system stays in a safe state.
The Banker’s algorithm is the classic avoidance algorithm.
Detection and recovery: allow deadlocks, detect them with a wait-for graph, then recover by killing or rolling back a process.
Ignoring: many operating systems simply ignore the problem (the ostrich algorithm) because deadlocks are rare.